

Claude Prices Probability, Not Payoffs

Phase 1 Results: Structural Estimation of Risk Preferences in an LLM

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1 Introduction

This project treats large language models as economic agents and recovers their preference primitives from their choices. The full study elicits binary lottery choices from LLMs, estimates structural discrete-choice models on the resulting data, and then carries the identical instruments to matched base and instruction-tuned models to test whether post-training instills behavioral biases. Phase 1 is the deliberate warm-up: fit a constant-relative-risk-aversion (CRRA) expected utility model to 1,200 controlled choices from a single frontier model, and in doing so debug the elicitation pipeline before the stakes rise. The purpose of a warm-up is to be easy. This one was not, and the way it failed is the finding.

The model under study is `claude-haiku-4-5-20251001`, a pinned snapshot, queried at temperature 1.0 through forced tool-choice with a 10% plain-text validation arm. We asked it 1,200 either/or money questions of the form “a sure \$86 or a 43% chance of \$210,” each one procedurally generated to be contamination-safe, fully counterbalanced for option order, and drawn under one of five paraphrase templates in a fresh context. A textbook CRRA fit returned a plausible risk-aversion number and hid a specification failure underneath it. The diagnostics that the design was built to run caught the failure. Once corrected, the model reveals a preference structure that is coherent, human-improbable in a specific and quantifiable way, and directly relevant to how agentic systems will behave when they allocate real resources.

2 The Result in Plain Language

We asked an AI model twelve hundred simple money questions. Every question had the same shape: a guaranteed amount of money, or a gamble. For example, “Would you rather take a sure \$86, or a 43% chance at \$210 and otherwise nothing?” We varied the guaranteed amount, the size of the prize, and the odds of winning across forty distinct gambles, asked each one many times with different wording and different ordering, and recorded which side the model picked. Nothing was left to chance in the design: the numbers were odd and non-round so the model could not be reciting a memorized textbook, and every gamble appeared in both possible orders so we could tell whether mere position was steering the answer.

A person deciding these questions weighs two things: how much money is on the table, and how likely they are to win it. A long shot at a huge prize can be worth taking; a near-certain shot at a

small prize can be worth passing on. The model does something stranger. It looks almost entirely at the odds of winning and barely at the size of the prize.

Here is the behavior in one picture. Offer the model a gamble it will probably lose, and it refuses, no matter how large the prize. Raise the prize from \$200 to \$2,000 to \$20,000 while keeping the win probability low, and the refusal does not budge. Now offer it a gamble it will probably win, even one whose prize is only mediocre, and it takes it. When the chance of winning was below roughly 30%, the model gambled about 8% of the time. When the chance of winning was above roughly 60%, it gambled about 86% of the time. The average money-value of the gambles was nearly identical across those two groups. The model was not chasing value. It was chasing the likely winner.

This is close to the opposite of a well-known human quirk. People buy lottery tickets: they overvalue tiny chances at enormous prizes. This model would never buy a lottery ticket at any price, and it would happily take a coin-flip for pocket change. Neither is “rational” in the economist’s sense, and the model’s version is arguably the more consequential one for anything asked to make real decisions.

We found one more thing, and it is almost embarrassing in how simple it is. Merely listing the gamble second, rather than first, made the model much more likely to choose it. Same gamble, same numbers, same odds; only the position on the page changed, and the model’s choices shifted by a large and statistically overwhelming margin. Whoever controls the order in which options are presented has a lever on what this model decides.

Why care? Because these models are being handed real decisions: which supplier to pick, which trade to place, which case to prioritize, what to offer in a negotiation. A decision-maker that ignores how much is at stake and fixates on the odds will systematically walk past high-value long shots (the research bet, the insurance-like tradeoff) and grab safe mediocrity. That is a predictable, exploitable distortion, and Phase 1 measures it.

3 Experimental Design

Every choice in the study is an independent API call with no memory of prior trials. The instrument is a forced binary choice between a sure amount and a two-outcome gamble. The gamble pays a high amount with probability p and zero otherwise. The design holds to the measurement commitments the proposal set out, treating the known validity threats in this literature as first-order.

- **Contamination-safe gambles.** Every lottery is procedurally generated with non-round payoffs and jittered probabilities inside novel wording. There are no textbook Holt-Laury rows to recite.
- **Full counterbalancing of order.** Each gamble is presented in both orders, with the sure option and the gamble swapping first and second position, so position can be estimated rather than assumed away.
- **Template as a factor.** Each gamble is drawn under five semantically equivalent paraphrase templates, so between-template variance is observable.
- **Reproducibility.** Every trial logs the model snapshot, seed, temperature, and raw response. Elicitation is via forced tool-choice, with a 10% plain-text arm to validate that the tool-choice channel is not itself distorting answers.

The design grid is 40 gambles (sure amount \$23–\$187 against a two-outcome gamble with high payoff at probability $p \in [0.19, 0.83]$, else zero; EV ratios spanning $0.75\text{--}1.7) \times 5$ templates $\times 2$ orders $\times 3$ repetitions, for 1,200 target choices. The final dataset contains 1,200 valid choices with zero unparseable responses. A mid-run API credit outage caused 265 trials to fail; these were re-run cleanly, and because run order was shuffled, the outage is random with respect to experimental conditions.

4 Econometric Model and Estimation

4.1 Baseline: CRRA expected utility with a Fechner error (M1)

The canonical structural exercise models the model as a CRRA expected-utility maximizer with a logit (Fechner) choice error. Utility is

$$u(x) = \frac{x^{1-r}}{1-r},$$

and the probability of choosing the gamble over the sure option is

$$\Pr(\text{gamble}) = \frac{1}{1 + \exp(- (EU_g - EU_s)/(\mu \cdot \text{scale}))},$$

where $EU_g = p \cdot u(\text{hi})$, $EU_s = u(\text{sure})$, and $\text{scale} = |u(\text{max payoff})|$ normalizes the noise term across stakes. Maximum likelihood on the 1,200 choices returns $r = 0.19$ (se 0.048, 95% CI $[0.10, 0.29]$) and $\mu = 0.26$ (se 0.040), with log-likelihood -809.69 . Read on its own, this is an unremarkable, mildly risk-averse agent.

4.2 Diagnostic: probability drives choice, payoffs do not

The sanity checks the design was built to run flag M1 as misspecified. The share of gambles chosen is non-monotone across EV-ratio quartiles (0.43, 0.71, 0.45, 0.62); there is a large position gap of 0.12 ($p < 0.001$); and the spread across templates is 0.25. A descriptive logistic regression of the gamble choice on the EV ratio, the win probability p , and a `safe_first` indicator (the gamble listed second) makes the pattern explicit:

$$\text{ev_ratio: } +1.71 \ (z = 5.0), \quad p: +11.87 \ (z = 22.8), \quad \text{safe_first: } +0.97 \ (z = 6.3).$$

Holding the mean EV ratio roughly constant (between 1.08 and 1.19) across win-probability bins, uptake climbs from 8.3% to 85.9% as p rises (Table 3). Choice tracks the probability of winning; payoff magnitudes barely register. CRRA cannot generate this: it integrates probability and payoff into a single expected utility, so it cannot make choice nearly independent of payoff while sharply dependent on p .

4.3 Extended model: Prelec weighting plus a position parameter (M2)

The diagnostic points to a distorted probability weighting function. M2 adds a Prelec (1998) weighting function and a position nuisance parameter δ that enters the latent index when the gamble is listed second:

$$w(p) = \exp(-(-\ln p)^\gamma), \quad z = \frac{w(p) u(\text{hi}) - u(\text{sure})}{\mu \cdot \text{scale}} + \delta \cdot \mathbf{1}[\text{gamble second}],$$

with $\Pr(\text{gamble}) = 1/(1+e^{-z})$. At $\gamma = 1$ and $\delta = 0$ this nests M1 exactly, so the two are comparable by a likelihood-ratio test. Estimation is by maximum likelihood with multi-start Nelder-Mead and standard errors from the numerical Hessian.

5 Results

Table 1 reports the M2 estimates. Table 2 compares the baseline, the extended model, and a bare-bones probability-only logit (M3). Table 3 shows fit against the observed choice rate by win-probability bin.

Table 1: M2 parameter estimates ($n = 1,200$).

Parameter	Estimate (se)	Interpretation
r (curvature)	0.592 (0.028)	Risk aversion; human-plausible range
γ (Prelec)	2.693 (0.166)	Extreme S-shape; long shots underweighted
μ (Fechner noise)	0.104 (0.008)	Choice noise, less than half of M1's
δ (position)	1.257 (0.160)	Shift toward the gamble when it is listed second

Table 2: Model comparison. Lower AIC/BIC is better.

Model	Structure	k	logLik	AIC	BIC
M1	CRRA + Fechner	2	-809.69	1623.4	1633.6
M2	CRRA + Prelec + position	4	-501.24	1010.5	1030.8
M3	Probability-only logit	2	-501.40	1010.8	1021.0

The likelihood-ratio test of M1 against M2 is decisive: $\chi^2(2) = 616.9$, $p = 1.1 \times 10^{-134}$. Adding probability weighting and a position term transforms the fit.

Table 3: Choice rate by win-probability bin: observed vs. fitted. EV ratio is roughly constant across bins, so a value-integrating model should be near-flat.

Win probability p	Observed	M1 fitted	M2 fitted
(0.15, 0.30]	0.083	0.427	0.028
(0.30, 0.45]	0.138	0.474	0.161
(0.45, 0.60]	0.607	0.493	0.654
(0.60, 0.85]	0.859	0.487	0.782

M1 predicts a nearly flat 0.43–0.49 across bins, missing the observed climb entirely. M2 tracks the step from near-zero uptake at low p to high uptake at high p .

6 Interpretation

The weighting function is extreme, and points the wrong way. A Prelec $\gamma = 2.69$ is a strong S-shape. It severely underweights low and moderate probabilities and only credits near-certain odds. Humans show the opposite: an inverse-S with γ around 0.65–0.75 (Prelec 1998; Wu

and Gonzalez 1996), overweighting long shots, which is why people buy lottery tickets. Haiku radically underweights anything short of a likely win. It will not take a long shot at nearly any price.

Once weighting is controlled, the curvature is ordinary. The M2 estimate $r = 0.59$ sits squarely in the standard human CRRA range. The baseline $r = 0.19$ was misspecification bias: forcing a probability-distortion pattern through a pure-curvature model pulled the curvature estimate toward risk neutrality. The apparent risk attitude in M1 was an artifact of the wrong model.

Position moves choice by a large factor. A position parameter $\delta = 1.26$ implies an odds shift of $e^{1.26} \approx 3.5\times$ toward the gamble merely from listing it second. The direction is notable: this is a second-option bias, opposite the first-option “A-bias” that much of the LLM literature reports.

Noise was mostly misspecification. The Fechner noise term halved, from $\mu = 0.26$ in M1 to $\mu = 0.10$ in M2. What looked like random choice under the wrong model was largely structure the wrong model could not represent.

The honest caveat. The probability-only heuristic M3 essentially ties M2 on AIC (1010.8 vs. 1010.5) and beats it on BIC (1021.0 vs. 1030.8). Win probability alone explains almost everything; payoff magnitudes add little. This near-tie is itself the finding. The model behaves closer to a “pick the likely winner” heuristic than to genuine expected-utility integration. The structural model earns its place by interpretability: it nests the rational benchmark, yields parameters comparable to the human behavioral literature, and locates exactly where and how the model departs from that benchmark, which a bare logit cannot do.

7 Policy Relevance

Allocative agents inherit this distortion. LLMs are being deployed to make allocative choices: procurement, trading, triage, negotiation. An agent that thresholds on probability and ignores payoff magnitude will systematically decline high-value long shots (R&D bets, insurance-like tradeoffs) and accept likely-but-mediocre options. Phase 1 quantifies that distortion for one snapshot under one protocol, turning a plausible worry into a measured parameter.

Interface order is a control lever. A $3.5\times$ position effect means whoever controls the order in which options are presented steers the agent’s choice. Interface design becomes a lever over AI decisions. That matters for anyone building or regulating agentic systems, and it matters more in adversarial settings, where an opponent who arranges the menu can tilt the outcome.

Structural estimation earns its keep. A descriptive bias demonstration cannot separate “the model is risk-averse” from “the model ignores payoffs.” The M1 fit would have reported the former; the truth is the latter. Structural estimation with a correctly specified model can tell them apart, and the misspecification was only detectable because the design carried sanity checks and counterbalancing. This is the argument for measurement-grade evaluations rather than one-off bias anecdotes.

Scope. These parameters are properties of this pinned snapshot under this protocol: unincin-
tivated, forced-choice, one model. They are not claimed as universal traits of LLMs. Phase 2 (gain/loss framing, loss aversion, an endogenous reference point) and Phase 3 (base versus instruct attribution) test whether the pattern is stable and where it comes from.

8 Limitations and Next Phase

Phase 1 studies a single model, at a single temperature, with unincentivized forced choice, so the estimates are conditional on that protocol. The near-tie between the structural model and the probability-only heuristic means the payoff-magnitude channel is weak in this design; a stakes range wider than \$23–\$187 might surface more curvature, and Phase 2 will widen it. The position effect is estimated as a single nuisance term rather than modeled by template, though the counterbalanced design ensures it does not bias the preference parameters. Temperature 1.0 sampling is treated as a choice error process here; whether that mapping holds is a separate question the logprob elicitation planned for Phase 3 will address directly.

Phase 2 adds the gain/loss frame manipulation and estimates loss aversion and an endogenous reference point. Phase 3 runs the identical instruments on matched base and instruction-tuned open-weight models to attribute the estimated parameters to post-training. Phase 1 has done its job: the pipeline is validated end to end, and it has already returned a sharp, quantified, and policy-relevant result.

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